

Oluwatobi Melvyn Mayungbo

Software Engineer, Payments and Risk Systems

melvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

SUMMARY

Backend and AI engineer who builds Python APIs that stay correct under load. Built a payments ledger API with FastAPI, PostgreSQL, Redis and Alembic, idempotency keys and a transactional outbox, a hybrid retrieval assistant with guardrails, and an MCP server with caching, rate limits and retries. Deploys with Docker and GitHub Actions.

SKILLS

Engineering and MLOps: Python, FastAPI, PostgreSQL, Redis, Docker, CI/CD (GitHub Actions), Pytest, GCP Cloud Run, AWS ECS Fargate, SQL, DuckDB, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit, React

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Model Context Protocol (MCP), Prompt Engineering, Citation Verification, Hugging Face, ONNX Runtime

Payments and Risk: Double-Entry Ledgers and Idempotency, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

PROJECTS

Production-Grade Double-Entry Ledger API | [GitHub](#)

Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.
- Required an Idempotency-Key on every mutating endpoint so a replayed key returns the original response and creates nothing; events go out through a transactional outbox.
- Load tested to find throughput saturating at about 250 to 260 req/s; removing the hot account cut transfer p99 from 1800 ms to 390 ms (4.6×) at 50 users.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.
- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.

- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.
- Telco Customer Churn, End to End on AWS** | [GitHub](#)
Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch
- Served it through a FastAPI /predict endpoint with a Gradio UI, containerised with Docker and deployed on AWS ECS Fargate behind an Application Load Balancer, with CloudWatch logs and GitHub Actions CI/CD.

EXPERIENCE

- Machine Learning Instructor, SQI College of ICT** March 2026 to Present
- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan** January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action